

PRASANNA KUMAR

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PROFESSIONAL SUMMARY

Computer Science Engineering student (2022–2026) specialising in Artificial Intelligence, Machine Learning, Generative AI, and Data Science. Built two production-grade AI systems: a Multimodal EV AI Diagnostic Platform (LLMs, RAG, Computer Vision, Agentic AI) and a Deep Learning Deepfake Detection system (CNN, Vision, Transformer, FastAPI). Skilled in Python, RAG, Prompt Engineering, Embeddings, Semantic Search, Vector Databases, Conversational AI, Agentic AI, and full-stack AI deployment. Completed Deloitte Data Analytics Job Simulation (Forage) and an AI/ML internship at Vision Astraa EV Academy. Seeking internships and entry-level roles in Data Science, AI/ML Engineering, Generative AI, Agentic AI, and Data Analysis.

PROFESSIONAL EXPERIENCE

AI/ML Data Science Intern | Vision Astraa EV Academy, Bangalore

Feb 2026 – May 2026

- Completed structured AI/ML for EV internship (VTU curriculum requirement; Cert ID: EVA/2526/2225) applying machine learning and data science to real-world EV datasets.
- Performed exploratory data analysis (EDA), data preprocessing, data cleaning, data wrangling, and feature engineering on structured EV telemetry data using Python, Pandas, and NumPy.
- Built and evaluated predictive models (regression, classification) using Scikit-learn; applied model training, model evaluation, and hyperparameter tuning techniques.
- Created data visualization reports and dashboards using Power BI and Matplotlib to communicate statistical analysis and insights to engineering stakeholders.
- Contributed to data pipeline and ETL workflows for automated data ingestion; produced KPI reporting and business intelligence summaries for EV analytics.

TECHNICAL SKILLS

Programming Languages: Python, SQL, HTML, CSS, JavaScript

AI & Machine Learning: Machine Learning, Deep Learning, Neural Networks, Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, Embeddings, Semantic Search, NLP, Computer Vision, Conversational AI, Agentic AI, AI Agents, Multi-Agent Systems, Tool Use, Function Calling, Workflow Automation, Fine-Tuning, Transfer Learning, Model Training, Model Deployment, Inference

GenAI & LLM Stack: Ollama, Phi-3, LLaVA, Vector Databases, FAISS, Sentence-Transformers, Multimodal AI, AI Orchestration

Frameworks & Libraries: Flask, FastAPI, REST API, Scikit-learn, Pandas, NumPy, OpenCV, Matplotlib, Seaborn, (ViT)

Development Tools: Git, GitHub, Jupyter Notebook, Google Colab, VS Code, Conda

Data & Analytics: EDA, Exploratory Data Analysis, Data Cleaning, Data Wrangling, Feature Engineering, Statistical Analysis, Statistics, Predictive Analytics, ETL, Data Pipeline, Business Intelligence, KPI Reporting, Dashboard Development, Data Visualization, Data Storytelling, Power BI

PROJECTS

EV AI Diagnostic Platform — Multimodal AI Copilot | *Python, Flask, Ollama, Phi-3, LLaVA, FAISS, Sentence-Transformers, OpenCV*

- Designed and deployed a production-grade Multimodal AI Diagnostic Copilot integrating Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Computer Vision, Telemetry Analytics, and Conversational AI for intelligent EV diagnostics and decision support.
- Built embeddings and semantic search pipelines using FAISS, Sentence-Transformers, and Vector Databases for context-aware EV knowledge retrieval and accurate question answering over technical manuals.
- Integrated Phi-3 and LLaVA Vision-Language Models with Prompt Engineering and AI Orchestration to enable multimodal reasoning, image analysis, and Agentic AI workflows with tool use and function calling patterns.
- Designed modular workflow automation pipelines: battery diagnostics, telemetry interpretation, charging analysis, range prediction, and engineering calculations — demonstrating multi-agent and AI agent architecture.

- Deployed Flask-based REST API with multimodal chat interface (text + image), scalable model inference, and GitHub deployment — evolved from RAG chatbot to a complete Agentic AI Copilot.

Deepfake Detection System | Python, FastAPI, Deep Learning, CNN, Vision Transformer (ViT), OpenCV, Xception, timm

- Built an end-to-end deep learning pipeline for deepfake video detection using a hybrid CNN + Vision Transformer (ViT) neural network architecture — demonstrating expertise in model training, transfer learning, and fine-tuning.
- Implemented automated face extraction and data pipeline from video frames using OpenCV with 80/20 train/test splits across four datasets: Actors, Deepfakes, FaceSwap, and DeepFake Detection.
- Trained DeepfakeDetector_v2 using Xception CNN backbone and ViT-Tiny (patch16-224); applied transfer learning, classification, and model evaluation metrics for robust binary classification of manipulated faces.
- Deployed a FastAPI web application with JWT authentication, video upload, inference endpoint, and real-time deepfake detection results — demonstrating full model deployment practices.
- Designed modular CLI structure (extract → train → test → start) with conda environment management; implemented statistical analysis and model evaluation for performance benchmarking.

EDUCATION

Bachelor of Engineering – Computer Science

2022 – 2026

HKBK College of Engineering, Bangalore | Visvesvaraya Technological University (VTU)

Pre-University Science (PCMB)

2019 – 2021

Infant Jesus PU College, Raichur, Karnataka

CERTIFICATIONS & TRAINING

Data Analytics Job Simulation — Deloitte Australia (via Forage)

July 2025

↳ Completed practical tasks in Data Analysis, Statistical Analysis, Data Visualization, and Forensic Technology in a simulated Deloitte consulting environment.

↳ Applied data storytelling, reporting, and business intelligence skills to real-world analytical scenarios.

Verification ID: 2nSaJiBFM5jsRZDLN | Issued by Forage

AI/ML for EV — Internship Completion — Vision Astraa EV Academy (VTU Affiliated) Feb – May 2026

↳ Certified completion of hands-on AI/ML project covering machine learning, data science, predictive analytics, and EV telemetry analysis.

↳ Issued by Program Coordinator and Co-Founder & CEO, Vision Astraa EV Academy.

Certificate ID: EVA/2526/2225

KEY ACHIEVEMENTS

- Independently built two production-grade AI systems — a Multimodal Agentic AI Copilot (LLMs, RAG, Embeddings, Computer Vision) and a Deepfake Detection system (CNN, ViT, model deployment) — while still in undergraduate studies.
- Implemented Agentic AI patterns including tool use, function calling, AI orchestration, and workflow automation from scratch — demonstrating strong architectural understanding of multi-agent systems.
- Earned Deloitte Data Analytics Job Simulation certificate (Forage), demonstrating industry-standard data analysis, statistical analysis, and business intelligence skills.
- Covered both Generative AI (LLMs, RAG, Multimodal AI, Agentic AI) and Discriminative AI (Deep Learning, CNN, ViT, Computer Vision) — demonstrating full breadth across the AI and Data Science landscape.

ADDITIONAL INFORMATION

Soft Skills: Communication, Team Collaboration, Adaptability, Problem-Solving, Critical Thinking, Self-directed Learning

Languages: Kannada (Native), English (Professional), Telugu (Fluent), Hindi (Fluent)

Date of Birth: 04 March 2003

Portfolio: prasannakumar9i.github.io